INFORMATION SYSTEMS INVOLVED:
- Messaging System (used for service communications)
- Ingestion System (collect records from client-side systems, analyze records/samples flow, not the sa
- Preparation System (preprocess samples, correct samples values, extract features)
- Segregation System (collect prepared sessions and create learning sets)
- Development System (train, validate, test the classifier)
- Production System (the classifier)
- Evaluation System (periodically evaluates the performances of the classifier)
- Client-side systems (provide various input records)

BPMN MODELING
- for each process, create the handoff level first, and then modify it as a service level workflow
- by default, the task type is script
- an asterisk denotes a parameter to configure

Process CONFIGURE SYSTEMS ("human-glue" version)
1 configuration received (Messaging system, human)
2.a.1 configure (Ingestion System, human) → 3
2.b.1 configure (Preparation System, human) → 3
2.c.1 configure (Segregation System, human) → 3
2.d.1 configure (Development System, human) → 3
2.e.1 configure (Production System, human) → 3
2.f.1 configure (Evaluation System, human) → 3
2.g.1 configure (client-side systems)
3 wait for the end of 2.a, 2.b, …, 2.g (Messaging system, human)
3 configuration completed (Messaging system, human)

Process PREPARE SESSION
(Ingestion System)
1 record received
2 store record
3 if insufficient record (*) then END
4 create raw session
5 remove records
6 mark missing samples
7 if raw session invalid (*) then END
8 if evaluation phase then send label to evaluation system
9 raw session sent (*)

(Preparation System)
1 raw session received
2 correct missing samples
3 detect and correct absolute outliers (*)
4 extract features (*)
5 if development phase then prepared session sent to segregation system (*)
   else prepared session sent to production system (*)

Process GENERATE LEARNING SETS
(Segregation system)
1 prepared session received
2 store prepared session
3 if sessions not sufficient (*) then END
4 generate balancing report
5 check data balancing (human,*)
6 if unbalanced classes then END configuration sent (*)
7 generate coverage report
8 check coverage (human, *)

9 if coverage not satisfied then END configuration sent (*)
10 generate learning sets (*)
11 learning sets sent

Process DEVELOP CLASSIFIER
(Development System)
1 learning set received
2 set average hyperparameters (*)
3 set #iterations (human)
4 train
5 if ongoing validation then go to 9
6 generate learning report
7 check learning plot (human)
8 if # iterations not fine then go to 3
9 set hyperparameters
10 if ongoing validation then go to 4
11 generate validation report
12 check validation results (human, *)
13 if there is no valid classifier then go to 2
14 generate test report
15 check test results (human, *)
16 if test not passed then END configuration sent (*)
      else END classifier sent (*)

(production system)
1 classifier received
2 deploy classifier
3 configuration sent(*)

Process CLASSIFY SESSION
(production system)
1 prepared session received
2 classify
3 if evaluation phase then send label to evaluation system
4 label sent (*)

Process EVALUATE CLASSIFIER PERFORMANCE
(Evaluation system)
1 label received
2 store label
3 if #labels not sufficient (*) then END
4 generate evaluation report
5 remove labels
6 evaluate classifier (human, *)
7 if classifier is not good then END configuration sent (*)
   else END